Areas Related to Circles (Set B) — MCQ Worksheet

Mathematics • Chapter 12 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. Area of a circle of radius 21 cm (π = 22/7):

(A) 1386 cm^2 (B) 693 cm^2
(C) 132 cm^2 (D) 1320 cm^2

Q2. Circumference of a circle of radius 7 cm (π = 22/7):

(A) 44 cm (B) 22 cm
(C) 88 cm (D) 11 cm

Q3. Area of a sector with angle 60° and radius 6 cm (π = 22/7):

(A) 132/7 cm^2 (B) 44/7 cm^2
(C) 12 cm^2 (D) 22 cm^2

Q4. Length of arc of sector 90° in circle of radius 14 cm (π = 22/7):

(A) 22 cm (B) 11 cm
(C) 44 cm (D) 88 cm

Q5. Area of a quadrant of a circle of radius 14 cm (π = 22/7):

(A) 154 cm^2 (B) 77 cm^2
(C) 44 cm^2 (D) 22 cm^2

Q6. If diameter of a circle is 14 cm, area is (π = 22/7):

(A) 154 cm^2 (B) 44 cm^2
(C) 22 cm^2 (D) 77 cm^2

Q7. Radius of a circle whose area is 1386 cm^2 (π = 22/7):

(A) 7 cm (B) 14 cm
(C) 21 cm (D) 11 cm

Q8. Area of an annulus (ring) between two concentric circles of radii 7 cm and 4 cm is:

(A) 33π cm^2 (B) 11π cm^2
(C) 21π cm^2 (D) 49π cm^2

Q9. Length of pendulum which sweeps an area of 24 sq cm in 1/4 revolution:

(A) $\sqrt{(96/\pi)}$ cm (B) 8 cm
(C) 6 cm (D) 12 cm

Q10. Minute hand of a clock is 14 cm long. Distance it covers in 1 hour (π = 22/7):

(A) 88 cm (B) 44 cm
(C) 176 cm (D) 132 cm

Q11. Area swept by minute hand of length 14 cm in 5 minutes (π = 22/7):

(A) 154/3 cm^2 (B) 44 cm^2
(C) 77 cm^2 (D) 11 cm^2

Q12. A wheel diameter 42 cm. Distance covered in one revolution:

(A) 132 cm (B) 66 cm
(C) 264 cm (D) 44 cm

Q13. If circumference and area of a circle have same numerical value, radius is:

(A) 2 (B) 1
(C) π (D) 4

Q14. Ratio of area of sector with angle θ to area of circle is:

(A) θ/360° (B) 360°/θ
(C) θ/π (D) θ^2/360°

Q15. Area of segment of circle of radius 7 cm subtending angle 90° at centre is:

(A) $(77/2 - 24.5)$ cm^2 ≈ 14 cm^2
(B) 77/2 cm^2
(C) 44 cm^2
(D) 100 cm^2

Q16. A square is inscribed in a circle of radius r. Area of square is:
(A) r^2
(B) $2r^2$
(C) $4r^2$
(D) πr^2

Q17. A horse is tied at one corner of a 20 m × 25 m field by a rope 10 m long. Area it can graze ($\pi = 22/7$):
(A) 550/7 m^2 ≈ 78.5 m^2
(B) 100 m^2
(C) 300/7 m^2
(D) 176 m^2

Q18. The wheels of a car turn 1000 times to cover 660 m. Diameter of wheel is:
(A) 21 cm
(B) 42 cm
(C) 14 cm
(D) 70 cm

Q19. A bicycle wheel makes 5000 revolutions in moving 11 km. Diameter is ($\pi = 22/7$):
(A) 70 cm
(B) 140 cm
(C) 35 cm
(D) 10 cm

Q20. Area enclosed between two concentric circles is 770 cm^2. If radius of outer is 21 cm, inner radius is ($\pi = 22/7$):
(A) 14 cm
(B) 10 cm
(C) 7 cm
(D) 11 cm

Q21. Area of a circular ring is 88 cm^2. If the difference in radii is 1 cm, sum of radii is ($\pi = 22/7$):
(A) 14 cm
(B) 28 cm
(C) 7 cm
(D) 21 cm

Q22. A racetrack is in the form of a ring whose inner circumference is 352 m and outer is 396 m. Width of track ($\pi = 22/7$):
(A) 7 m
(B) 11 m
(C) 14 m
(D) 22 m

Q23. Two circles touch externally. The sum of their areas is 130π cm^2 and distance between centres is 14 cm. The radii are:
(A) 3 cm and 11 cm
(B) 5 cm and 9 cm
(C) 7 cm and 7 cm
(D) Cannot be determined

Q24. If a chord of a circle of radius 14 cm subtends 60° at centre, area of corresponding minor segment is ($\pi = 22/7$):
(A) $(308/3 - 49\sqrt{3})$ cm^2 ≈ 17.7 cm^2
(B) 100 cm^2
(C) 44 cm^2
(D) 50 cm^2

Q25. Area of an equilateral triangle inscribed in a circle of radius 6 cm is:
(A) $27\sqrt{3}$ cm^2
(B) $36\sqrt{3}$ cm^2
(C) $9\sqrt{3}$ cm^2
(D) $18\sqrt{3}$ cm^2

— End of Worksheet —